

require robust networking features, such as those deployed in large-scale IoT networks. Contiki supports a variety of wireless protocols and network topologies, making it a versatile choice for diverse IoT applications.

**FreeRTOS.** FreeRTOS is a widely used real-time operating system known for its performance and reliability. It offers a small memory footprint and deterministic behaviour, making it suitable for applications that require precise timing and resource optimisation. FreeRTOS is commonly used in embedded systems, including IoT devices, due to its efficiency and ease of use.

**Zephyr.** Zephyr is an open source, scalable operating system that supports a wide range of hardware platforms. It provides a modular architecture and a rich set of device drivers, making it easy to customise and adapt to different IoT applications. Zephyr emphasises security and reliability, making it suitable for critical IoT applications that require robust security measures.

**RIOT OS.** RIOT OS is a multi-threading operating system designed for resource-constrained devices. It supports various hardware platforms and communication protocols, making it a flexible choice for diverse IoT applications. RIOT OS offers a modular design, allowing developers to easily customise the OS to their specific needs.

**Mbed OS.** Mbed OS is a commercial operating system developed by Arm, focusing on security, connectivity, and ease of development. It provides a comprehensive set of APIs and tools for IoT development, making it a popular choice for commercial IoT applications. Mbed OS is well-suited for devices that require robust security and connectivity features, such as those

deployed in industrial and automotive environments.

## Selection criteria for an IoT OS

When selecting an IoT operating system, consider the following factors:

### Hardware compatibility.

- **Processor architecture.** Ensure the OS is compatible with the processor architecture of your target device (e.g., Arm, x86, RISC-V).
- **Memory constraints.** Consider the OS's memory footprint and its ability to operate efficiently on devices with limited memory.
- **Peripheral support.** Verify that the OS supports the specific peripherals required by your application (e.g., sensors, actuators, wireless modules).

### Application requirements.

- **Real-time capabilities.** If your application requires precise timing and responsiveness, choose an OS with strong real-time capabilities, such as FreeRTOS or Zephyr.
- **Energy efficiency.** For battery-powered devices, prioritise energy-efficient OSes like TinyOS or Contiki.
- **Security.** If security is a critical concern, consider OSes with robust security features, such as Mbed OS or Zephyr.
- **Connectivity.** Evaluate the OS's support for various wireless protocols (Wi-Fi, Bluetooth, Zigbee, etc) and its ability to connect to cloud platforms.

## Development support and community size

**Developer tools and libraries.** A strong developer community and a rich ecosystem of tools and libraries can significantly accelerate development.

### Documentation and tutorials.

Comprehensive documentation and tutorials can help you get started quickly and efficiently.

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“  
Branding is what people say about you when you are not in the room.”



-Jeff Bezos

